

REform presents

Hair Dryer

recover and reuse



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Visit the REform website to download and print other manuals available for different small domestic appliances.

Welcome!

This instructional guide is designed to incentivize users to maximize the utility of salvaged electronic and mechanical components from discarded domestic appliances found within the household. By disassembling, rebuilding and refurbishing perfectly functional components into minimalistic DIY outcomes served for distinct purposes you are actively preventing functional hardware from entering the waste stream and extending these components' lifespans.

Beyond its ecological value, this project also serves as a hands-on sandbox for exploring the foundations of electronics and components which rule our technological-based society. Through the integration of mechanical design, simple kinematics, and microelectronics, you will develop a practical understanding of electricity, smart systems, and the structural foundations of functional consumer products.

This specific guide focuses entirely on the repair, reuse, and refurbishment of a salvaged blade, a high-RPM motor component, and a mechanical toggle button harvested from a standard hairdryer.

The contents of this guide are structured across three escalating levels of difficulty, allowing you to push your skills to your desired extent:

LEVEL 1

Create a mechanical design of a manually rotating ventilator with controlled velocity states.

LEVEL 2

Using/ combining different blades into the same module. Adding an external power source to obtain higher motor speed.

LEVEL 3

Designing an air purifier with designated HEPA filters.

Essentials

for a circular Design approach

Sustainable Maker Spaces

Maker spaces consist of collaborative engineering environments which provide digital fabrication tools to manufacture new structures, stepping away from mass production and focusing on localized design, also making use of scrapped materials.

Makerspace Delft

- Community workshop carried by experts
- Many types of machinery, ex. Laser Cutters and 3D printers



TU/e Innovation Space

- Multidisciplinary hub
- Hands-on prototyping
- Engineering projects

TU/e Vertigo

- Can gather scrap materials for projects



Maarktplaats

- Digital platform
- Can buy/sell components from online sellers or creators
- Users can filter their search to maximize sustainable alternatives



Repair Café Network

- Localized hubs across The Netherlands
- Keep devices operational
- Stores 'organ donor boxes' filled with salvaged components

Community repair and Exchange

Public spaces designed for individual makers and everyday citizens to donate, swap, and acquire salvaged physical electronic, mechanical parts and tools.

Companies providing sustainable component purchase
Accessible commercial component stores which supply modular development boards that can be infinitely reprogrammed over their lifecycle.

Sustainable Enterprises & Circular Businesses

These social enterprises evaluate devices for component buyback and structural recovery before moving to melting or raw shredding stages.

By executing these high-tier R-strategies (Reuse and Refurbish), this intervention closes the inner loop of the circular economy.

It can be inspiring to meet people involved in circular habits and approaches, and learn more about circular design and economy in exhibitions or conferences.



32-RC (Netherlands)

- Active surplus electronics buyer
- Offers used components back into the market

BlueCity (Rotterdam)

- Cross-Industrial business accelerator of startups
- Dedicated to turn waste streams into circular economic assets
- Hosts exhibitions, conferences.



Holland Recycling (Boxtel)

- IT hardware buyback
- Component recovery of functioning parts



Mirec (Eindhoven)

- Circular collection of surplus inventories
- 'Parts recovery' program to reclaim working parts from electronic waste streams

TinyTronics (Netherlands)

- Webshop for Dutch makers
- Supply standardized microelectronics which can be reused for many projects.



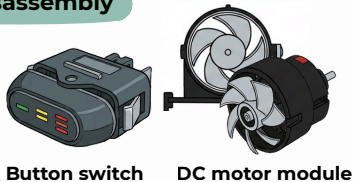
Adafruit Industries

- International hardware manufacturer
- Open-source microelectronics with accessible, repairable board layouts, limiting premature product obsolescence.



Inventory

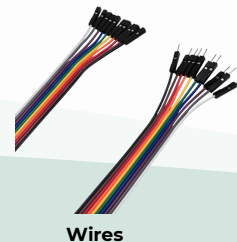
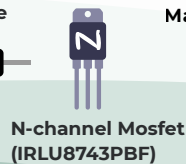
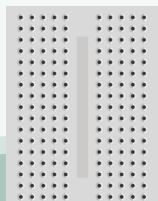
From Disassembly



Laser cut



170-pin Breadboard



Wire channel (2)



Wire attachment



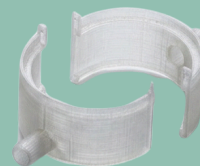
Wheel thread



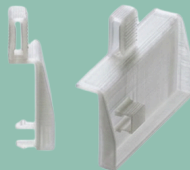
Button shaft



Motor casing



Clip-on



Main casing



Screws



Drawer slit



3D printed

Wire channel (1)



Electronics

Disassembly

Approaching a circular Design



Read More

Right to Repair

Every year, the global economy generates tens of millions of metric tons of electronic waste, a trajectory driven by planned obsolescence and unrepairable consumer product architectures.

The Right to Repair is a consumer and environmental movement advocating for the legal right of individuals to freely maintain, modify, and repair their own electronics, appliances, and vehicles. It incentivizes companies to design with a different perspective, competing against planned obsolescence, and facilitating consumers to dismantle smart products in a systematic and simple manner.

This act addresses small electronics through market-driven transparency incentives. For instance, In regions like the European Union, manufacturers are increasingly driven to calculate and display a mandatory Repairability Index score on their packaging. Nevertheless, companies are still given the freedom to follow their own design and manufacturing process, but this act is growing stronger by the day and influencing market strategies and brand reputation.

Mandatory Disassembly* & Electrical Safety Protocols

Whether experienced or not, when exploring with salvaged components it is crucial to understand and adhere to safety regulations to mitigate any possible risk:

- Isolation of Mains Voltage: Never initiate tool contact or case disassembly on any appliance that is physically connected or plugged into the AC grid. Disconnect the device entirely from the wall socket.
- Mechanical Isolation: During your experimentation, prioritize component integrity. Use the needed tools to open cases or separate parts instead of cutting or attempting to break these pieces, making them impossible to re-assemble.
- Eye Protection: Always wear safety eyewear for protection against any flying bits or material failures.

*Depending on the appliance, it is important to follow a safe and structured procedure. Search online for the specific hair dryer you own and find the precise disassembly instructions. Each company might follow a different procedure for their exact appliance.

Level 1

Page 9

Level 2

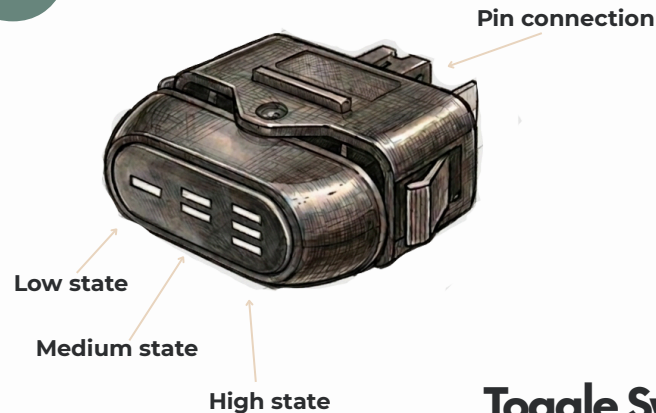
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Level 3

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Level 1

Overview Of Parts

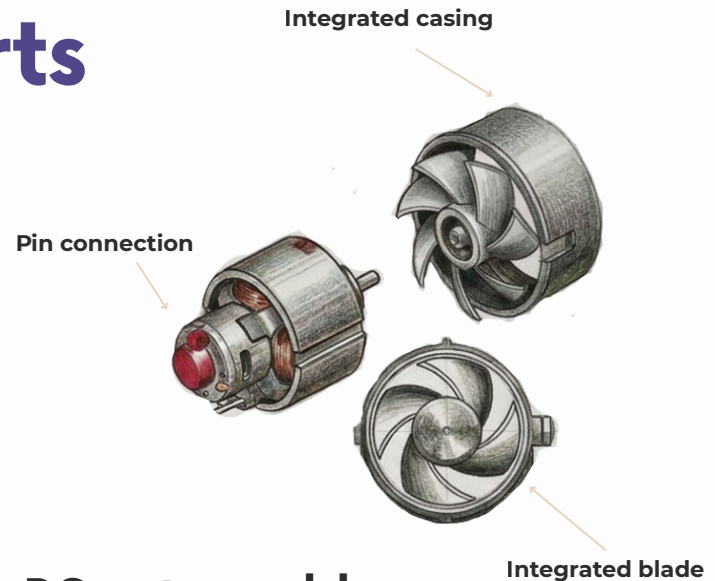


Toggle Switch

Function: This switch acts as a gatekeeper for the circuit, changing how electricity is routed through its three internal states. Red wire acts as the common power input line.

Pins:

- State 0 (LOW): Open circuit. 0 input, meaning the system is completely OFF.
- State 1 (MEDIUM): Bridges the internal connection between the Red and Yellow wires to initiate a low-speed circuit.
- State 2 (HIGH): Bridges the internal contacts completely, closing both the Red + Yellow loop and the Blue + Red loop simultaneously for maximum speed.

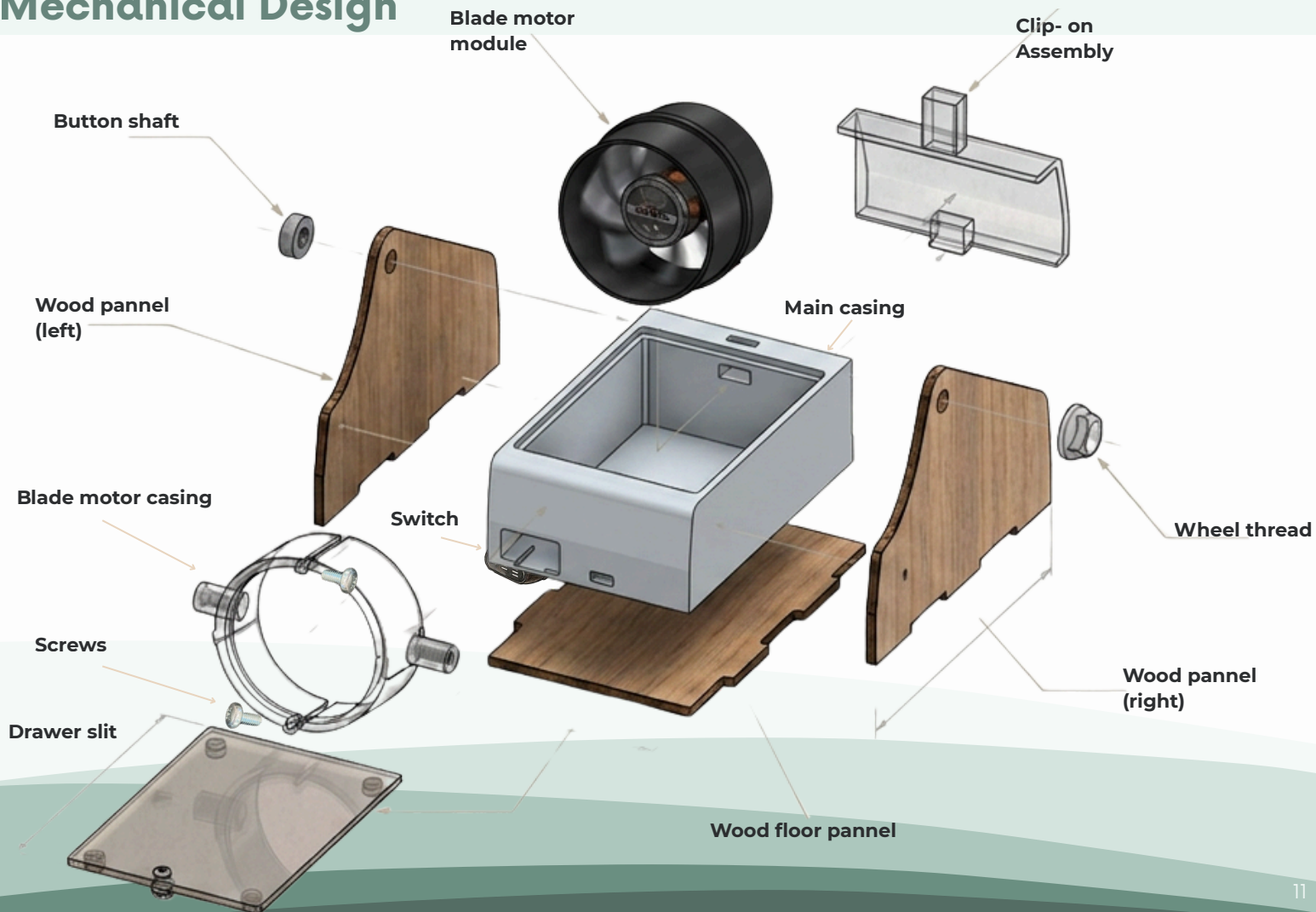


DC motor module

Function: When electricity passes through the motor's internal copper coils, it creates a magnetic field that spins the central rotor. This rotational force drives the attached blade, pulling air from the rear and driving it forward.

Pins: The motor features two terminal contacts. Depending on the positive and negative connections the motor will spin clockwise or anticlockwise.

Mechanical Design



1 Digital Fabrication

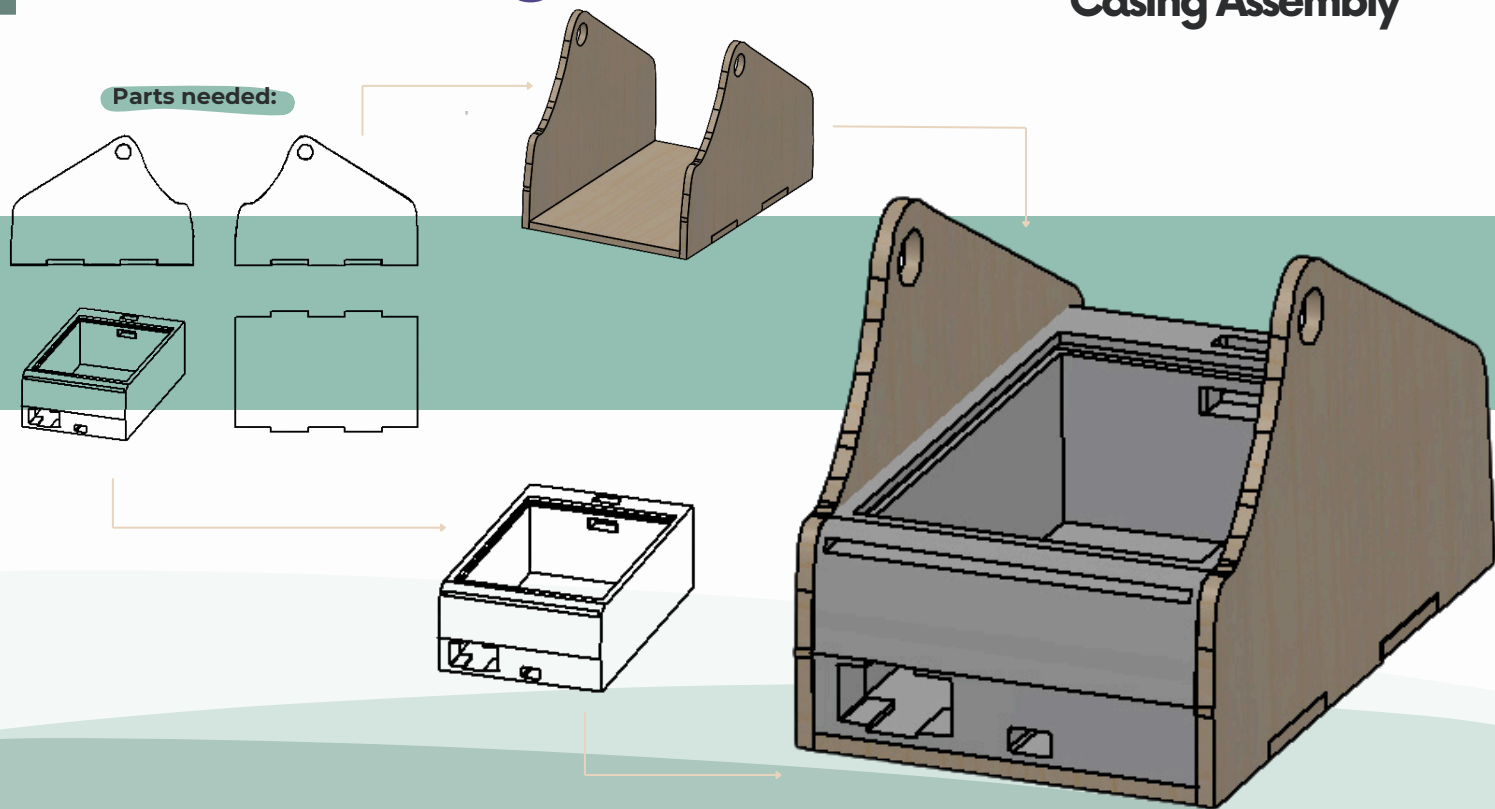
Before beginning structural assembly, access the REform digital website to retrieve these components. Download the optimized CAD files directly from our platform:

- 3D Printing: Download the .STL mesh files to print the two-part motor casing, the clip on attachment assembly, the casing, the wheel and the button shaft.
- Laser Cutting: Download the .DXF vector files to precision-cut the flat structural skeletal ribs from your chosen wood material. Dimensions are included so wood cutting can also be done.

If you want to dive deeper into the fundamentals of additive manufacturing and laser slicing, or if you want to attempt to design and customize these physical files yourself from scratch, follow our advanced prototyping guidelines on the REform website.

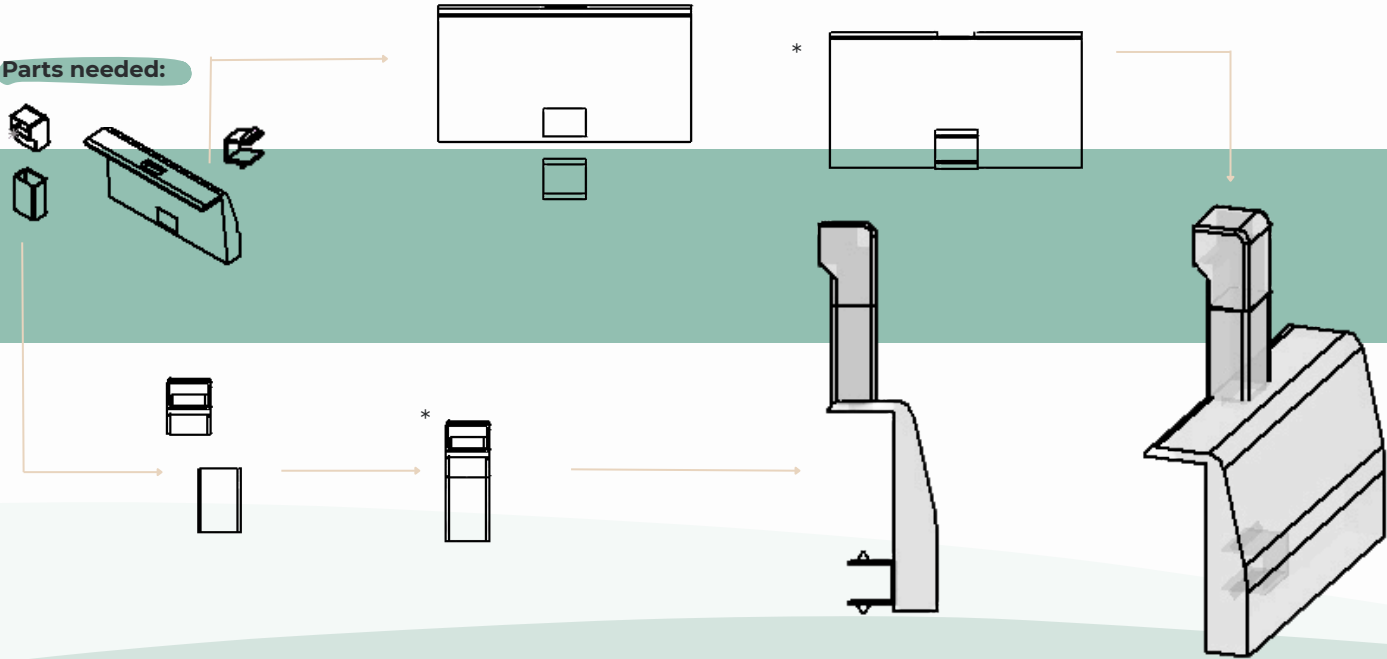
2 Assembling case

Casing Assembly



Clip-on Assembly

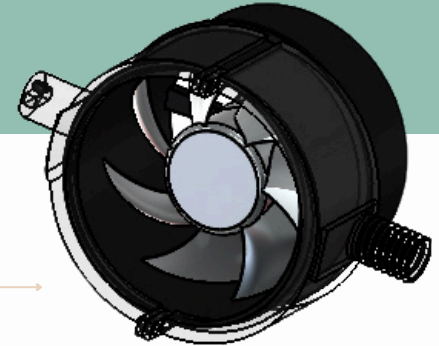
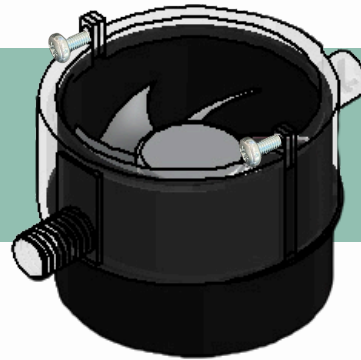
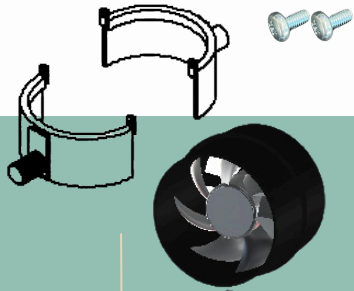
Parts needed:



*Due to the scope of this project these stages currently require permanent glue. This would not be the case in future iterations.

Motor casing Assembly

Parts needed:



Gather the needed screws (M3X5) as shown before, gather the left and right blade casing along with the motor blade component and screw the parts together as shown.

3 Overview Of Parts

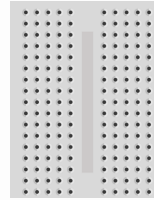


Resistors 10kOhm



ESP32 DEVKIT MODULE

1X



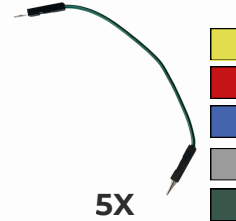
170-pin Breadboard



Diode



N-channel Mosfet (IRL...)



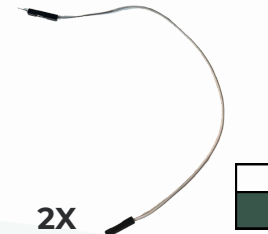
Wires, 10cm, male to male



Magnetic pogo pins (3 pin or 3 pin)



Wires 10cm, male to female



Wires 20cm, male to female

5 Wiring the Loop

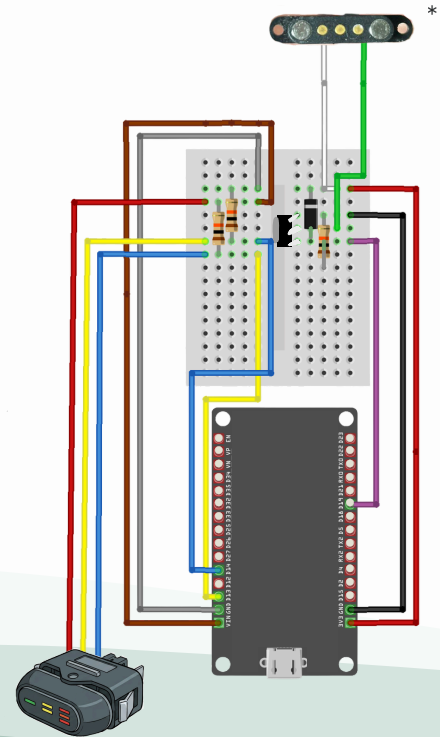
LEFT SIDE: Button Switch

1. **3V3** → **E4**
Brown wire, 10cm
2. **GND** → **E3**
Grey wire, 10cm
3. **GPIO 4** → **E7**
Yellow wire, 10cm
4. **GPIO 15** → **E6**
Blue wire, 10cm
5. **10K Resistor (1)** → **C3 and C7**
6. **10K Resistor (2)** → **B3 and B6**
7. **Button (R)** → **A4**
Red wire from button
8. **Button (B)** → **A6**
Blue wire from button
9. **Button (Y)** → **A7**
Yellow wire from button

RIGHT SIDE: Motor

1. **VIN** → **E3**
Red wire, 10cm
2. **GND** → **E5**
Black wire, 10cm
3. **GPIO 32** → **E7**
Purple wire, 10cm
4. **MosFet** → **A5, A6, A7**
5. **10K resistor (3)** → **C5 and C7**
6. **Diode B3 and B6**
7. **Motor wire** → **D3**
White wire, 20cm
8. **Motor wire** → **D5**
Green wire, 20cm

1. place the ESP32 upside down and below the breadboard, vertically aligning them together
2. Each side is for each component connection, it does not matter which side is set up first.



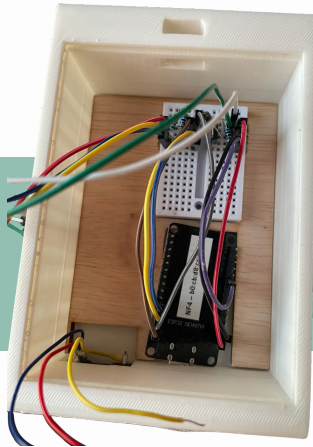
*For the scope of this project wires have been soldered to the magnetic pogo pins and motor blade, this will not be the case in future iterations.

6 Programming

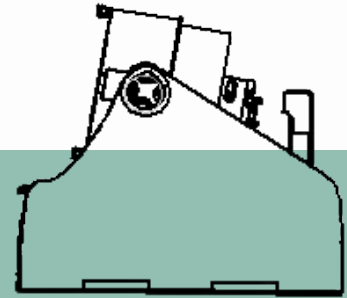
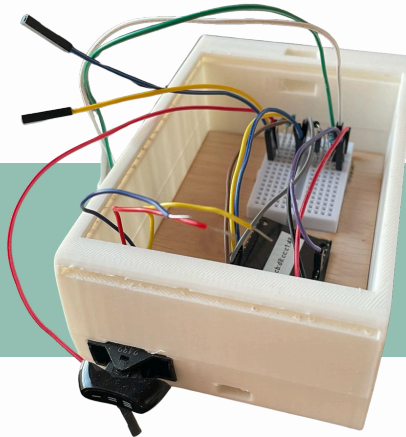
1. Install Arduino IDE on your computer
2. Go to REform to download our code for this exact electronic circuit
3. Plug the ESP32 fully wired and compile the code.



7 Final Setup

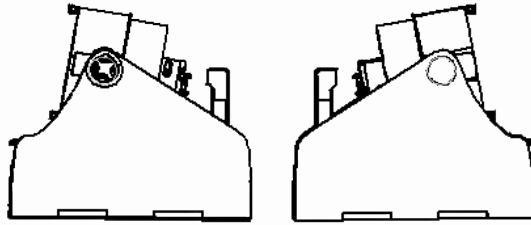


1. Place the electronics in designated laser cut support, located inside the casing. Then place the button in the casing space.



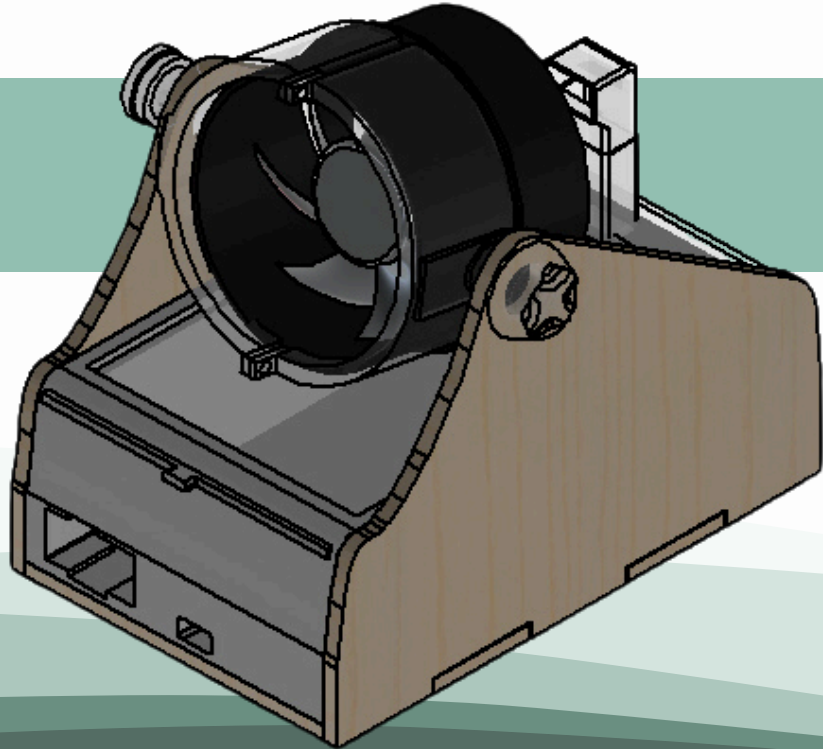
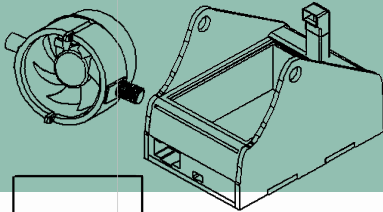
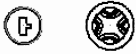
2. Pass the pogo pin wires through the wire channel and casing gap- leave the pogo pin outside the wire channel, visibly.

Assembly



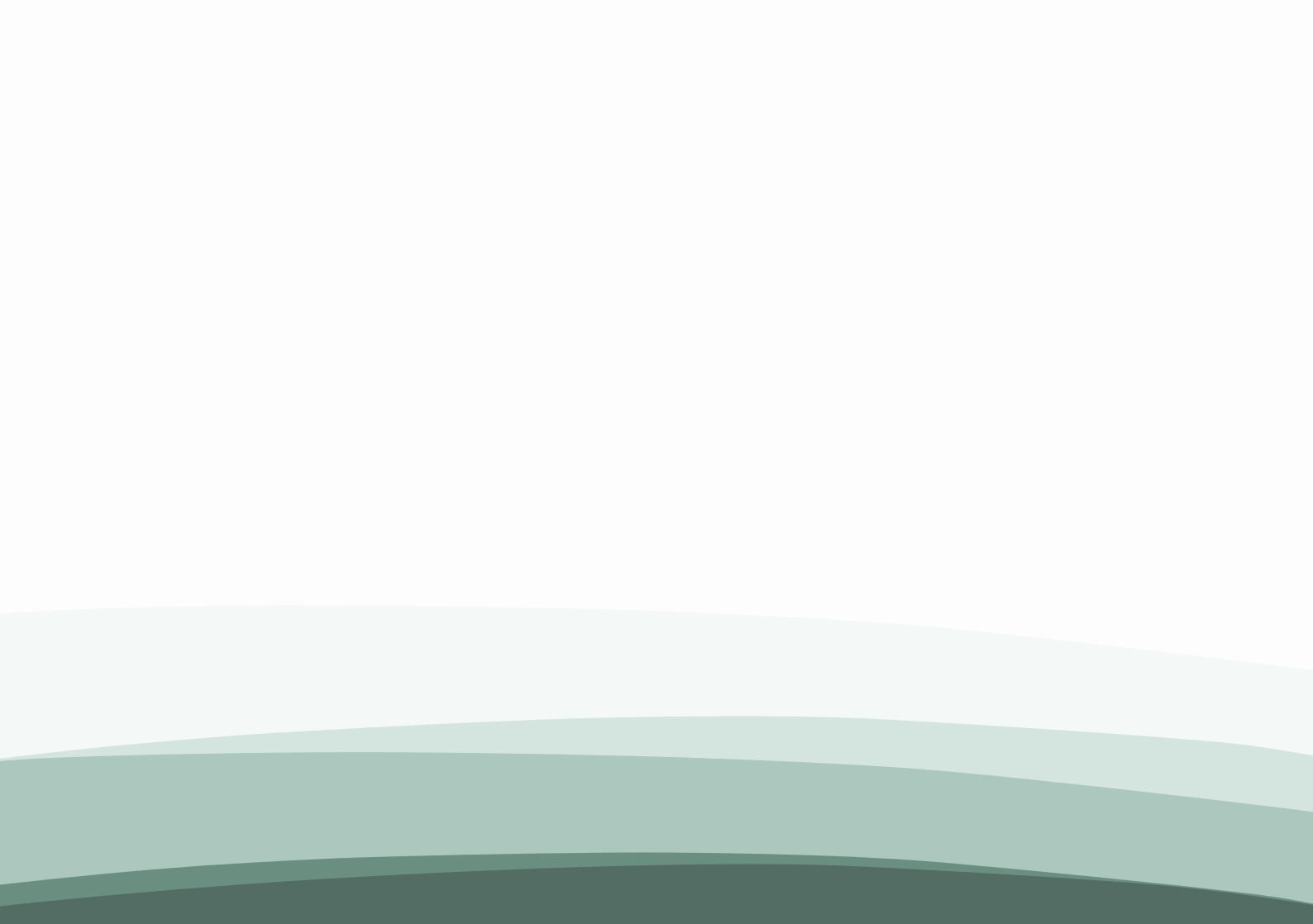
1. Insert the motor casing on the wood panel holes, insert and tighten the nut on one side.
2. Add the fixed button on the other side of the casing.
3. Plug the Laptop to the ESP32 and run the code. Motor should begin functioning along with the button states.

Parts needed:



Level 2

Level 3



REform